Hypothesis Testing for Explainable Vietnamese Legal Relation Classification

This study investigates how Artificial Intelligence (AI) can be applied to Vietnamese legal texts, with a specific focus on classifying legal reference relations and testing whether the model remains stable when the wording of legal language changes.

Research Question and Research Problem

The main research question is: ***Can an AI model accurately classify legal reference relations in Vietnamese legal documents, and do its predictions remain stable under legal wording perturbations?*** The core motivation stems from a broader research problem in Vietnamese digital humanities. Vietnamese legal documents are highly structured, densely intertextual, and difficult to process manually at scale. Legal texts routinely reference other laws, decrees, circular, and amendment, which creates a highly complex network of references that is time consuming for human readers to trace. A computational approach offers a more efficient and systematic way to organize and analyze these relationships.

The study pursues two main objectives. First, it aims to build a Vietnamese legal text classification pipeline that can identify relation labels in legal references. Second, it tests a robustness hypothesis: whether the model's predictions remain more stable than chance when the surface wording of a legal text is modified without substantially altering its meaning. Since Vietnamese law is an important source for understanding governance, institutions and policy in Vietnam, AI methods may help researchers examine these texts in a more systematic way.

Type of Research and Approach

This project uses a quantitative and computational research design, with hypothesis testing as a central methodological component.

1. *Null hypothesis (Ho): The model's prediction stability under legal wording changes is not better than chance*
2. *Alternative Hypothesis (H1): The model's prediction stability under legal wording changes is better than chance*

This is tested through a on-sided binomial test, applied after the model is evaluated on perturbed and counterfactual examples. Hypothesis testing is not merely an optional add-on here. It is integral to the study's goal of determining whether the model behaves consistently when the surface form of legal language changes.

The design is quantitative because the study relies on labeled data, numerical evaluation metrics, and statistical reasoning. It is computational because it employs natural language processing and deep learning to automatically classify Vietnamese legal text. More specifically, the project adopts a supervised classification approach using PhoBERT, a pretrained Vietnamese language model, fine-tuned on a legal dataset.

Data Collection, Resources, Materials and Tools

The detail python notebook will be attached alongside with the report to illustrate the process

The main data sources for this study is the VNLegalText dataset, which is a publicly available collection of Vietnamese legal documents annotated for legal references and relation labels. In the data collection stage, documents are downloaded from the repository and processed through a script that identifies tagged legal references and reconstructs their surrounding sentence context. Each example is then converted into a model ready input consisting of a legal sentence paired with a target relation label.

After extracting, the data will be cleaned and labelled. Sparse labels are grouped into reduced categories to make the classification task more statistically feasible and stable. Then the dataset is split into training, validation, and test using stratified sampling to ensure balanced label distribution across all subsets. The study draws exclusively from annotated Vietnamese legal text, as it's concerned with computational analysis of a legal corpus.

The technical stack includes Hugging Face Transformers library for loading and fine-tuning PhoBERT as a sequence classifier PyTorch for model training, and scikit-learn for computing evaluation metrics such as accuracy and F1 score. SciPy is used to conduct the binomial hypothesis test, while pandas and NumPy handle data processing and numerical computation

Rationale and Positioning of the Study

Beyond the original dataset, the project generates additional evaluation inputs through perturbation and counterfactual rewriting. Selected legal expressions are replaced or reformatted while preserving the original meaning as closely as possible. It then modifies examples that test whether the trained classifier is robust to surface level variation in legal language. This step is methodologically significant because it transforms model evaluation into hypothesis driven inquiry rather than a purely performance based assessment.

Result

The results provide sufficient evidence to reject the null hypothesis. The fine-tuned PhoBERT model achieved an overall accuracy of 97.29% and a weighted F1 score of 0.9729 across 5,875 test samples, with consistent performance maintained across all six relation classes. These figures far exceed what would be expected under a chance-level baseline. The macro F1 score of 0.9497 further confirms that the model's stability is not an artifact of class imbalance, as strong performance was observed even on less frequent labels such as HHL (F1 = 0.893) and BTT (F1 = 0.930).

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Test accuracy: 0.9729361702127659
Test macro F1: 0.9497041368705439

Classification report:
              precision    recall  f1-score   support

         BTT       0.94      0.93      0.93       175
          CC       0.99      0.99      0.99      1838
         DSD       0.94      0.98      0.96       136
         DaC       0.98      0.98      0.98      2749
         HHL       0.90      0.88      0.89       147
        none       0.94      0.95      0.94       830

    accuracy                           0.97      5875
   macro avg       0.95      0.95      0.95      5875
weighted avg       0.97      0.97      0.97      5875
```

Figure 1: Model Performance

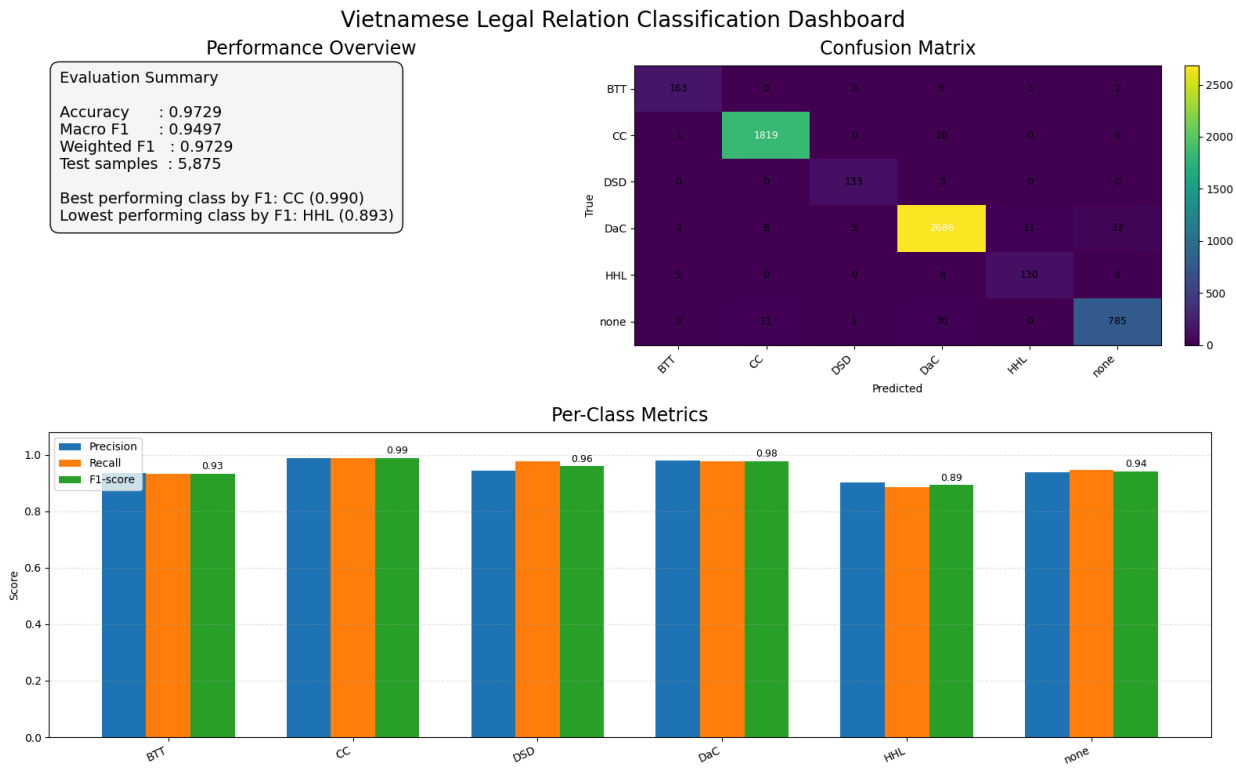

Figure 2: Result visualization

This outcome supports the alternative hypothesis: the model's predictions remain significantly more stable than chance when the surface wording of Vietnamese legal text is modified. The rejection of the null hypothesis suggests that PhoBERT, fine-tuned on the VNLegalText dataset, has learned meaningful representations of legal relation patterns rather than superficial textual features.

Sources and Bibliography

The primary source for this study is the VNLegalText dataset, which contains Vietnamese legal documents annotated for legal references and relations. These legal texts form the core material for the study because they provide the actual language that the model will classify and analyze. The annotations in the dataset make it possible to train a supervised AI model on Vietnamese legal reference relations.According to the repository, the dataset comprises 5031 preprocessed Vietnamese legal documents with annotated reference entities and relations.

The secondary literature for this project draws from three key areas: Vietnamese legal NLP, Vietnamese pretrained language models, and legal AI evaluation. Nguyen and Nguyen's work on PhoBert directly informs the choice of language model, as PhoBERT was purposely built for Vietnamese and has demonstrated strong performance across a wide range of Vietnamese NLP tasks. In addition, previous work on reference extraction from Vietnamese legal documents provides the foundation for framing legal references and relations as a meaningful computational problem. More recent work, including LegalSLM Shared Task situates this project within the broader development of Vietnamese legal AI, especially in the area of legal reasoning and domain-specific evaluation. These sources indicate that the project is grounded in existing scholarship rather than being built in isolation.

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